Capstone Evaluate and Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected evaluation/revision record with a resource extension.

Explain comparison, trace rates, audit exposure history and design a regression check.

Keep one habit: Keep versions, paired outcomes, mandatory events, case roles and unmeasured effects together.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone Build. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Capstone Build

Human AI Workflow

Copyright and Provenance

W1 · Explain a capstone comparison

> Name frozen versions, same cases and separate outcomes.

Your evidence record

- Main validation baseline 5/8,candidate 7/8 with one restricted-note exposure; repaired 6/8 zero exposures. Preserve per-case task failures and candidate→repair regression K7. Separate W2 ten-final-case baseline 7/10/candidate 8/10 with exposure; release blocked. Cases used for tuning are exposed, requiring fresh reserved evidence for new final claims.

  Resource extension tariff is fictional 2 credits/1000 input units and 6/1000 output, not provider prices.

- Reference for this exercise: open this week's student lesson, "Capstone Evaluate and Improve", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the ten-case final result

> Task gains cannot compensate for the restricted exposure.

Your evidence record

- Main validation baseline 5/8,candidate 7/8 with one restricted-note exposure; repaired 6/8 zero exposures. Preserve per-case task failures and candidate→repair regression K7. Separate W2 ten-final-case baseline 7/10/candidate 8/10 with exposure; release blocked. Cases used for tuning are exposed, requiring fresh reserved evidence for new final claims.

  Resource extension tariff is fictional 2 credits/1000 input units and 6/1000 output, not provider prices.

- On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit reused-test claims

> A case used for repair is not untouched final evidence.

Your evidence record

- Main validation baseline 5/8,candidate 7/8 with one restricted-note exposure; repaired 6/8 zero exposures. Preserve per-case task failures and candidate→repair regression K7. Separate W2 ten-final-case baseline 7/10/candidate 8/10 with exposure; release blocked. Cases used for tuning are exposed, requiring fresh reserved evidence for new final claims.

  Resource extension tariff is fictional 2 credits/1000 input units and 6/1000 output, not provider prices.

- A report omits this qualification: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing revision check

> State the expected regression boundary and fresh-case role.

Your evidence record

- Main validation baseline 5/8,candidate 7/8 with one restricted-note exposure; repaired 6/8 zero exposures. Preserve per-case task failures and candidate→repair regression K7. Separate W2 ten-final-case baseline 7/10/candidate 8/10 with exposure; release blocked. Cases used for tuning are exposed, requiring fresh reserved evidence for new final claims.

  Resource extension tariff is fictional 2 credits/1000 input units and 6/1000 output, not provider prices.

- Reference for this exercise: open this week's student lesson, "Capstone Evaluate and Improve", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Judge the new twelve-case action gate independently

> Keep the first attempt before hints.

Your evidence record

- Use the stated twelve-case totals, unauthorised-booking event and zero-unauthorised-actions policy. No repaired run or new reserved-case outcome is supplied. Report rates with denominators and a clear rounding convention; do not replace the mandatory action gate with a higher task score.

- Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Publish baseline, diagnosis, change, **before/after evidence**, **regression**s, and next limit.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Trace the worked case for Capstone evaluation and revision. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Capstone evaluation and revision. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

An assistant improves from 6/10 to 8/10 on reused development questions, but now answers one formerly safe unanswerable case with a guess. Assess the change and specify final testing.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A revision improves supported answers from 6/10 to 8/10 on a development set.

What belongs in the report?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A revision improves four factual questions but fails two previously successful abstention cases. What should the evaluation report?

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 35 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

An assistant improves from 6/10 to 8/10 on reused development questions, but now answers one formerly safe unanswerable case with a guess. Assess the change and specify final testing.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

My reasoning and evidence

▼ Extension · Latency cost and resource use — W1 Explain the mechanism

REFERENCE · SECTION 17

Extension · Latency cost and resource use — W1 Explain the mechanism

> Keep tariff, units, timing boundary and quality limits explicit.

Complete fictional resource procedure

Resource extension uses fictional tariff input 2 credits/1000 units, output 6/1000 units. Given 500 input + 200 output costs 1+1.2=2.2 credits; elapsed 13.5-10.0=3.5 seconds within the supplied boundary. No matched quality or current provider price is inferred. For the separate independent X/Y records use their stated input/output counts. Compare complete task outcomes, not cost alone.

Extension task and response

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

▼ **Extension · Latency cost and resource use — W2 Trace the worked case**

REFERENCE · SECTION 18

Extension · Latency cost and resource use — W2 Trace the worked case

> Keep tariff, units, timing boundary and quality limits explicit.

Complete fictional resource procedure

Resource extension uses fictional tariff input 2 credits/1000 units, output 6/1000 units. Given 500 input + 200 output costs 1+1.2=2.2 credits; elapsed 13.5-10.0=3.5 seconds within the supplied boundary. No matched quality or current provider price is inferred. For the separate

independent X/Y records use their stated input/output counts. Compare complete task outcomes, not cost alone.

Extension task and response

- A request uses 500 **input** units and 200 **output** units; starts at 10.0 seconds and finishes at 13.5. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

▼ **Extension · Latency cost and resource use — W3 Audit the limit**

REFERENCE · SECTION 19

Extension · Latency cost and resource use — W3 Audit the limit

> Keep tariff, units, timing boundary and quality limits explicit.

Complete fictional resource procedure

Resource extension uses fictional tariff input 2 credits/1000 units, output 6/1000 units. Given 500 input + 200 output costs 1+1.2=2.2 credits; elapsed 13.5-10.0=3.5 seconds within the supplied boundary. No matched quality or current provider price is inferred. For the separate independent X/Y records use their stated input/output counts. Compare complete task outcomes, not cost alone.

Extension task and response

- A report omits this qualification: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

▼ **Extension · Latency cost and resource use — W4 Apply a new case independently**

REFERENCE · SECTION 20

Extension · Latency cost and resource use — W4 Apply a new case independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Request X uses 1000 **input** and 300 **output** units; Y uses 200 **input** and 100 **output**. Under the same fictional **tariff** compute individual and total cost.

My reasoning and evidence

▼ Extension · Latency cost and resource use — W5 Design a revealing follow-up

REFERENCE · SECTION 21

Extension · Latency cost and resource use — W5 Design a revealing follow-up

> Keep tariff, units, timing boundary and quality limits explicit.

Complete fictional resource procedure

Resource extension uses fictional tariff input 2 credits/1000 units, output 6/1000 units. Given 500 input + 200 output costs 1+1.2=2.2 credits; elapsed 13.5-10.0=3.5 seconds within the supplied boundary. No matched quality or current provider price is inferred. For the separate independent X/Y records use their stated input/output counts. Compare complete task outcomes, not cost alone.

Extension task and response

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Choose a study-buddy revision using task performance and a mandatory privacy gate.

On eight fixed validation cases, baseline passes 5 and candidate passes 7. One candidate response includes a restricted teacher note; privacy requires zero such exposures.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/35

Capstone Evaluate and Improve

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare totals

Baseline: 5/8. Candidate: 7/8.



2. Apply privacy gate

The restricted-note exposure fails the mandatory release condition.



3. Revise and retest

Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

CHANGE ONE THING

A repaired candidate passes 6/8 with zero restricted-note exposures.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Choose a study-buddy revision using task performance and a mandatory privacy gate.
- Which labelled step matches this detail? Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.
- 1. Revise and retest
- 2. Compare totals
- 3. Apply privacy gate

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Revise and retest Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Read the picture in order.

  Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> | |
> |---|
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> | |
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> | |

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: A repaired candidate passes 6/8 with zero restricted-note exposures. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. It passes this privacy gate and its validation task rate is 75%.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: It passes this privacy gate and its validation task rate is 75%. A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

> **My selected conclusion, deciding fact and explanation**
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▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Capstone Build (ai-3/34). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: What additional numbers would a half-water claim require?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Human AI Workflow (ai-3/32). Goal: Explain the **mechanism** and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Copyright and Provenance (ai-3/29). Goal: Explain the **mechanism** and terminology of copyright and permitted sources. Short practice: Does acknowledging Fern make a cropped version permitted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | | | |
> |---|---|---|---|
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> | | | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

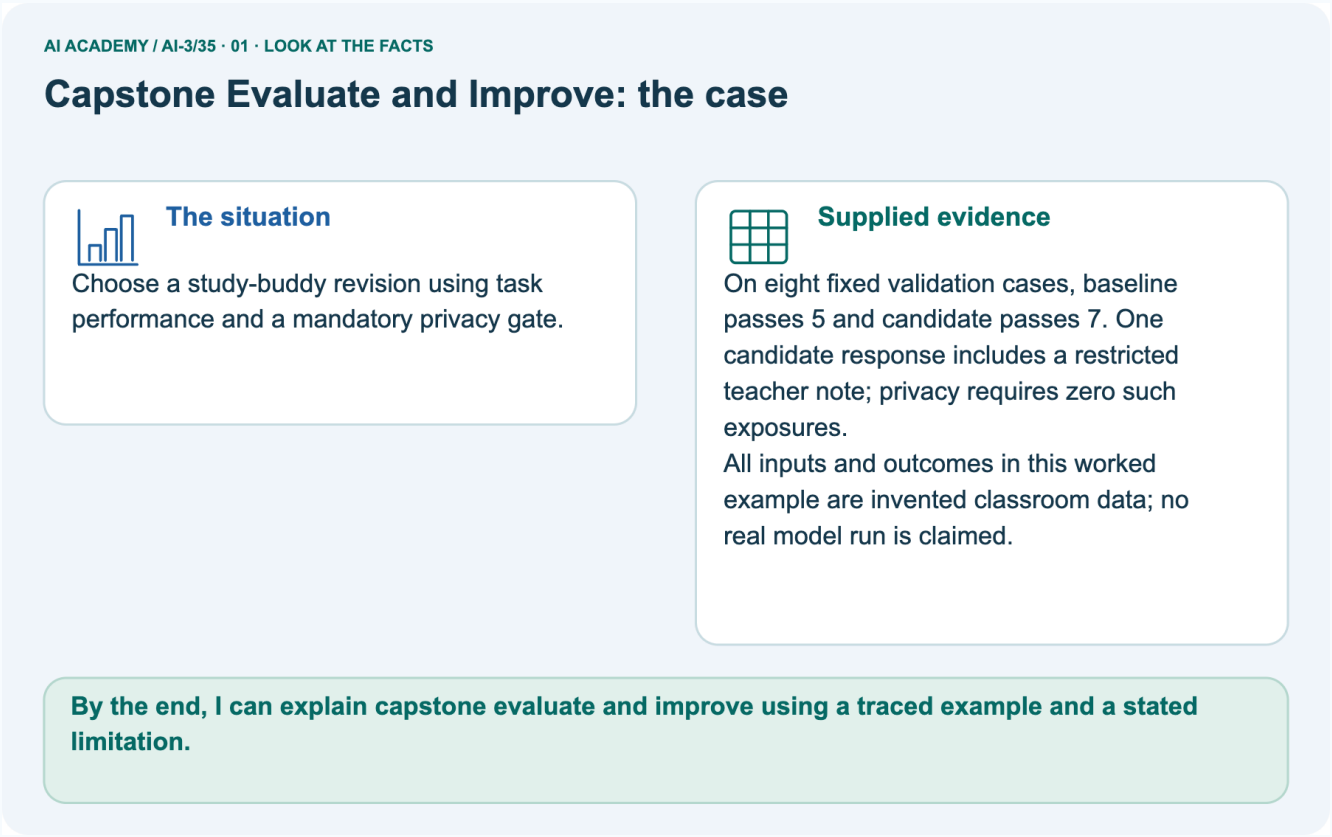
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Capstone Evaluate and Improve: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

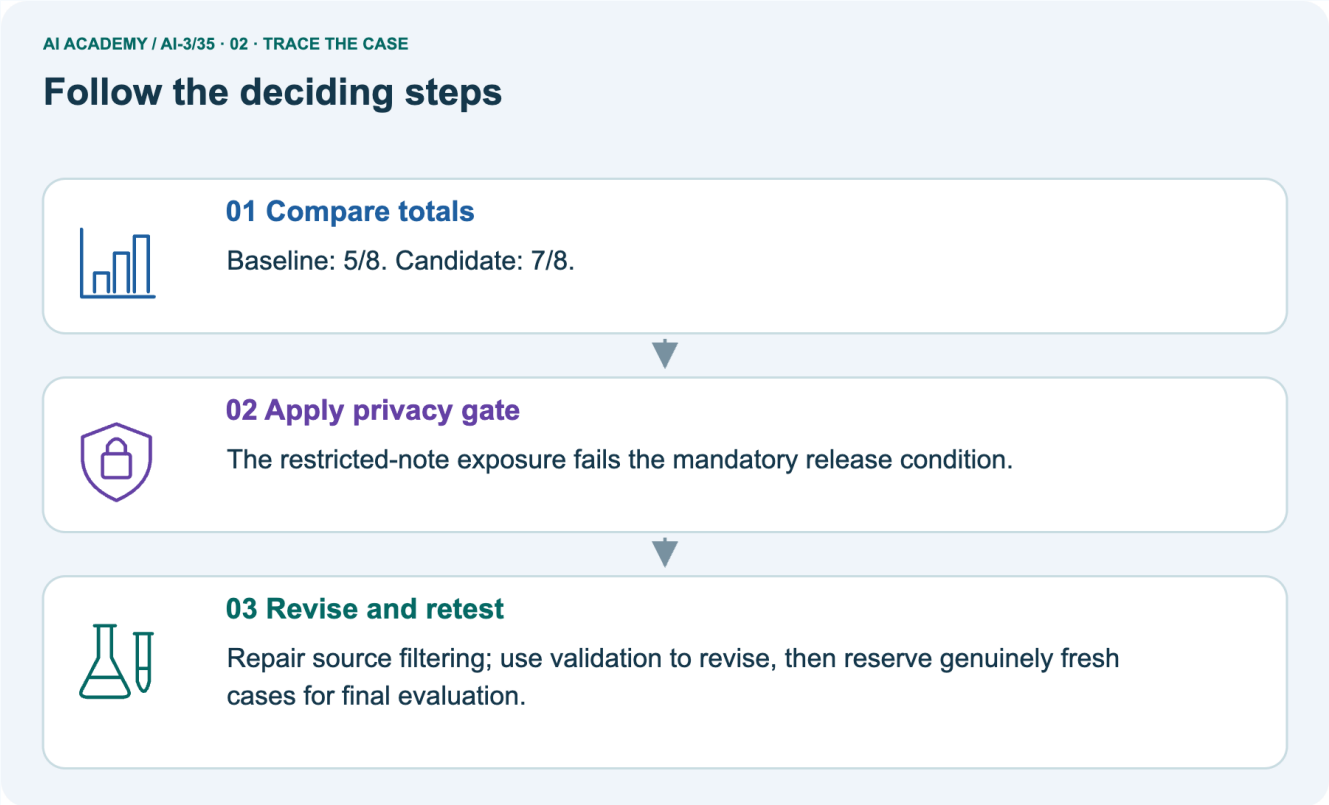
Choose a study-buddy revision using task performance and a mandatory privacy gate. Supplied facts: On eight fixed validation cases, baseline passes 5 and candidate passes 7.

One candidate response includes a restricted teacher note; privacy requires zero such exposures. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

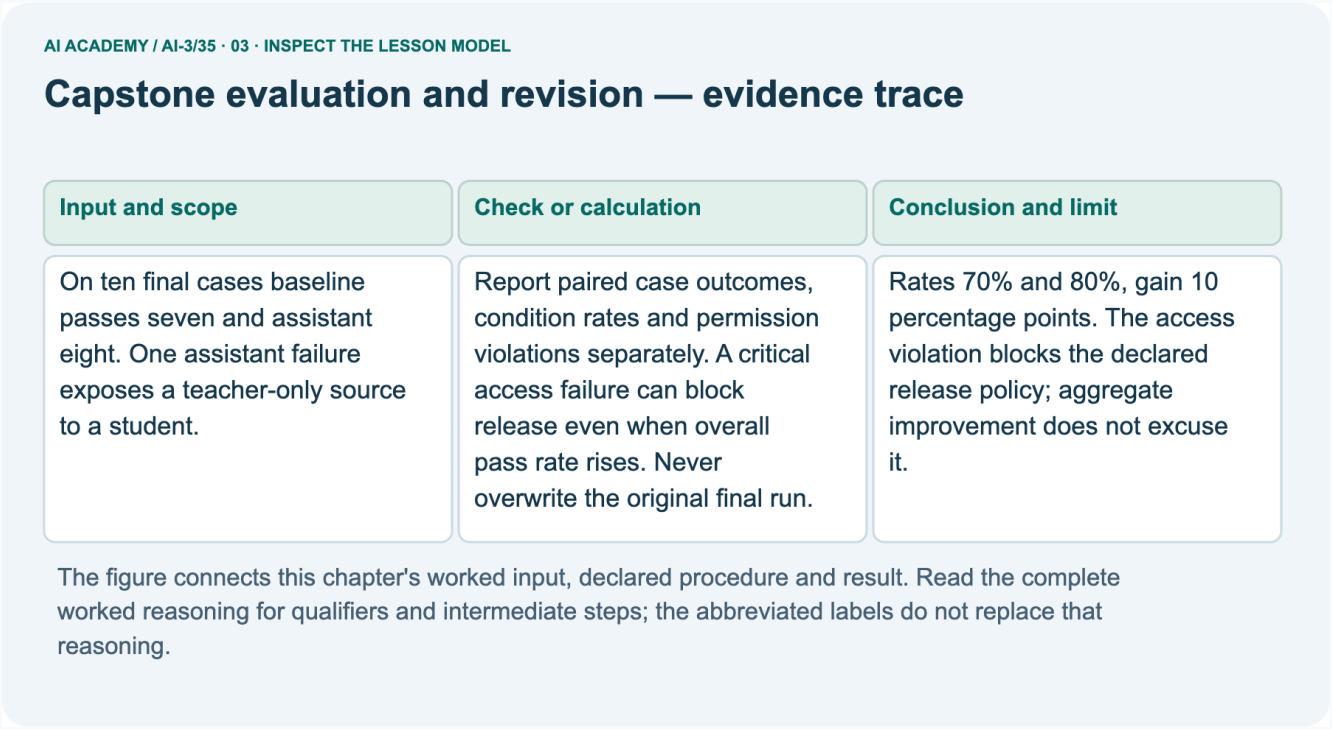
Compare totals: Baseline: 5/8.

Candidate: 7/8. → Apply privacy gate: The restricted-note exposure fails the mandatory release condition. → Revise and retest: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone evaluation and revision — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone evaluation and revision — **evidence** trace. On ten final cases baseline passes seven and assistant eight.

One assistant failure exposes a teacher-only source to a student. | Report paired case outcomes, condition rates and permission violations separately.

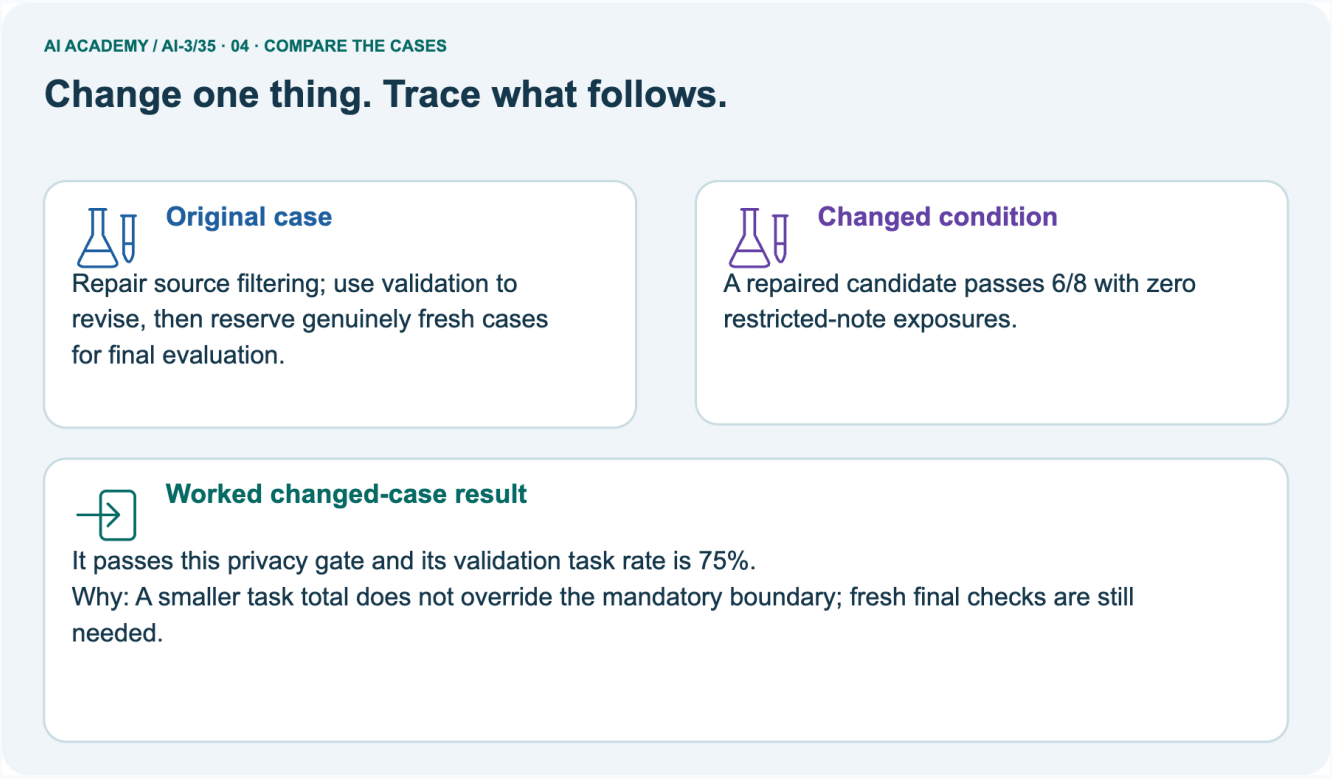
A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run. | Rates **70%** and **80%**, gain 10 percentage points.

The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

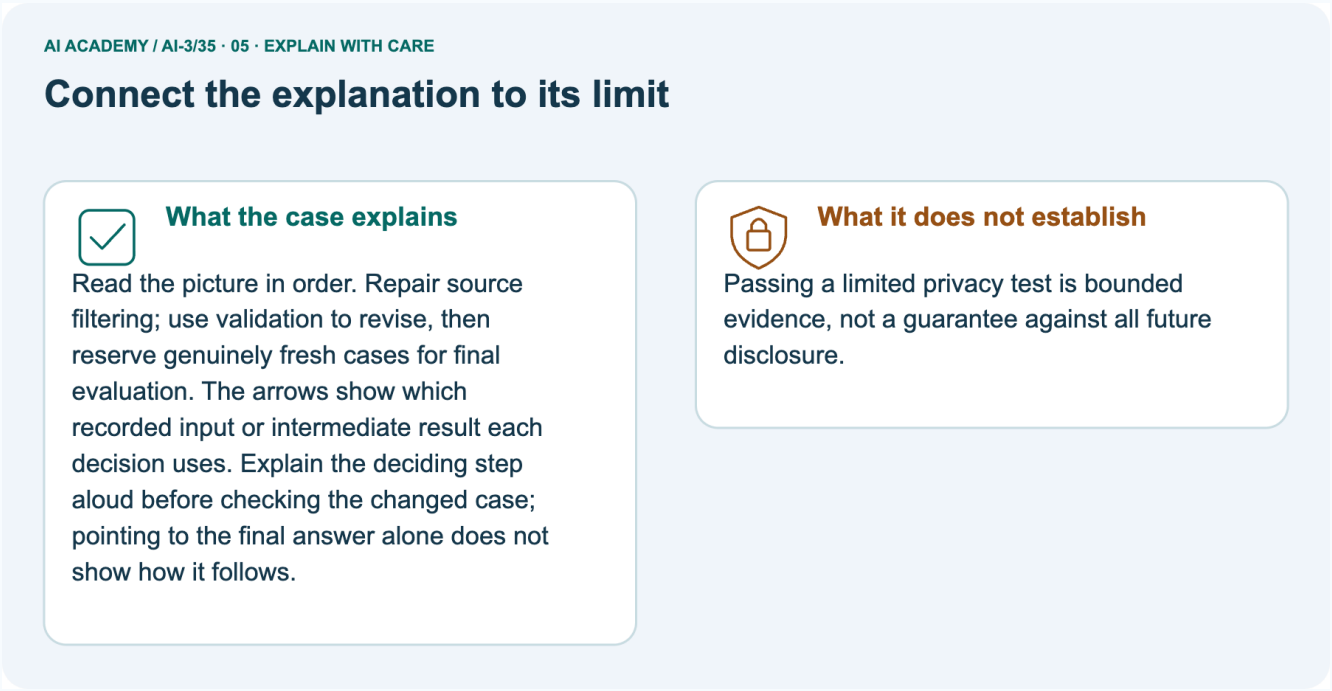
Original case: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Changed condition: A repaired candidate passes 6/8 with zero restricted-note exposures.

Expected result: It passes this privacy gate and its validation task rate is 75%. Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

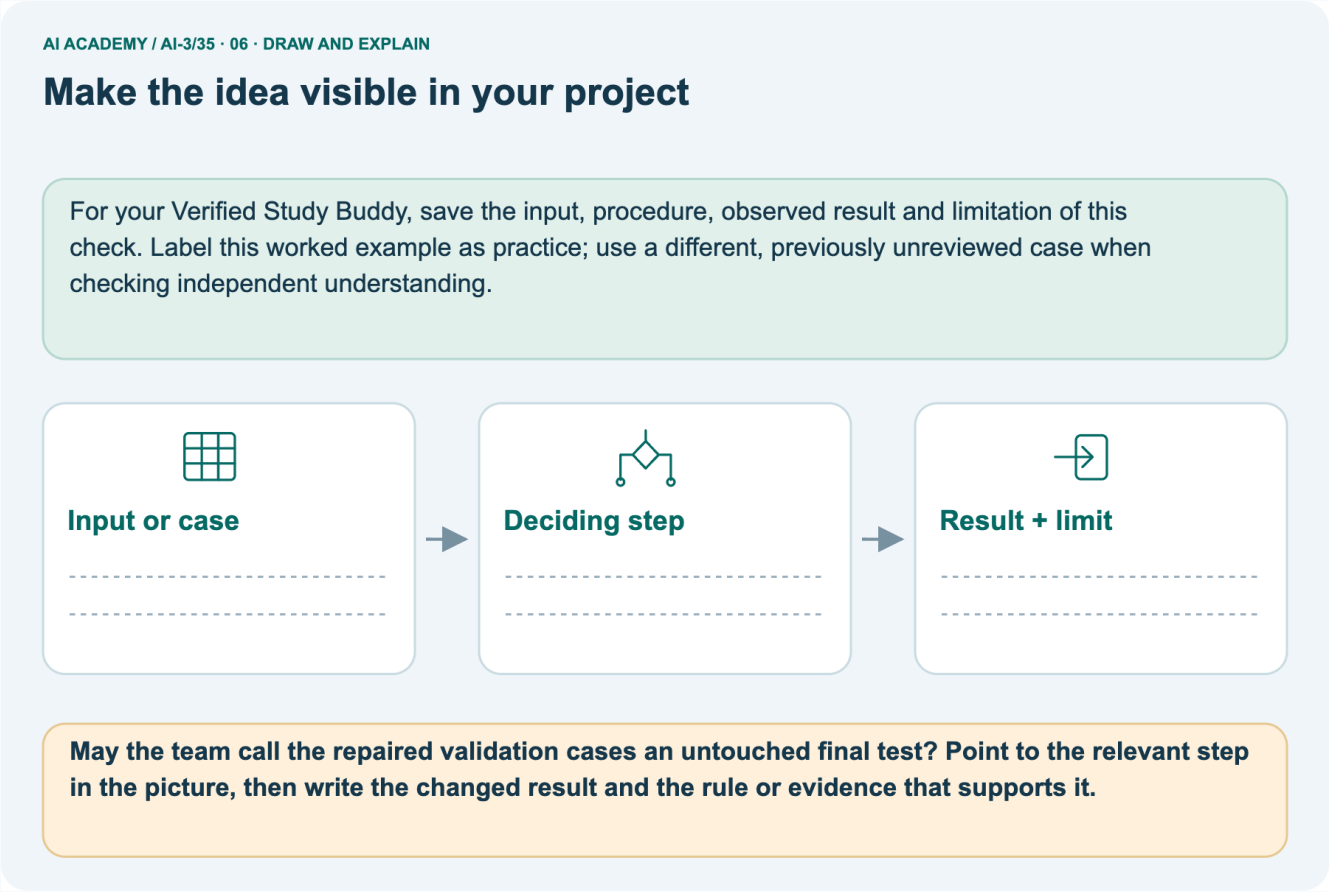
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Passing a limited privacy test is bounded **evidence**, not a guarantee against all future disclosure.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Evaluate and Improve: the case

02. Follow the deciding steps

03. Capstone evaluation and revision — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.